# Saverio **Monaco**

Computational Physicist · (Quantum) Machine Learning Researcher

*Hamburg, Germany*

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## Education

### RWTH Aachen University & DESY
***Hamburg, Germany***

PhD in Quantum Generative Models for High Energy Physics (ENGAGE) — *Mar. 2024 - Present*

- Marie Skłodowska-Curie PhD Fellowship
- Thesis: Detector Simulation and Jet Clustering for HL-LHC with Quantum Computing

### University of Padua
***Padua, Italy***

M.Sc. in Computational Physics — *Sep. 2020 - Sep. 2023*

- Thesis: Study of Quantum Correlations in LHCb Simulated Heavy Flavour Jets
- Honors: Magna cum laude

### University of Catania
***Catania, Italy***

B.Sc. in Physics — *Sep. 2016 - Jul. 2020*

- Thesis: Phase-Space Formulation of Quantum Mechanics

## Experience

### Research Secondment
***Hamburg, Germany***

ParityQC — *Sep. 2025 - Dec. 2025*

- Compared performance of training quantum generative models on IBM machines and the QSea machine.

### Quantum Machine Learning Intern
***Geneva, Switzerland***

CERN — *Apr. 2022 - Aug. 2022*

- Explored quantum machine learning techniques for phase detection in the ANNNI spin model.
- Developed and implemented supervised and unsupervised architectures for phase detection.
- Published a Python package under the CERN-IT organization.
- Authored and published a paper on the implemented techniques.

## Skills

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|---|---|
| **Quantum Computing** | PennyLane, Qiskit, Quimb, TeNPy, Yao.jl |
| **Programming Languages** | Python, Julia, C/C++, SQL, R, VHDL, Agda, LaTeX, Nix, JavaScript, Rust |
| **Machine Learning** | JAX, PyTorch, Keras, Optax, scikit-learn |
| **Other Libraries** | Pandas, NumPy, SciPy, Matplotlib, BeautifulSoup |
| **Tools** | Git, Docker, Vim, Linux, Sphinx, ReadTheDocs, Weights & Biases |
| **Languages** | Italian (native), English (C1, IELTS Academic 7.0), German (C1/B2, Goethe-Zertifikat B2), French (B2/B1, EsaBac Diploma) |

## Publications

**2026** Classical Pre-Training of Large Shallow Parametrized Quantum Circuits through Pauli Propagation and deployment on Quantum Hardware
S. **Monaco**, J. Slim, R. Xue, D. Krücker, K. Borras
*arXiv pre-print*

**2026** An IQP Born Machine for Calorimeter Image Generation at 64 Qubits with Compiled-IQP Deployment
J. Slim, S. **Monaco**, F. Rehm, D. Krücker, K. Borras
*arXiv pre-print*

**2026** Quantum Feature Amplification Network (QFAN) as An Autoregressive Quantum Generative Model
J. Slim, S. **Monaco**, F. Rehm, D. Krücker, K. Borras
*arXiv pre-print*

**2026** Interferometric Quantum Polynomial Chaos Expansion as a Generative Model for Calorimeter Shower Simulation
J. Slim, S. **Monaco**, F. Rehm, D. Krücker, F. Gaede, K. Borras
*arXiv pre-print*

**2026** A Coherent Memory Register for Sequential Quantum Generative Modeling, with Application to Calorimeter Showers
J. Slim, S. **Monaco**, R. Xue, D. Krücker, K. Borras
*arXiv pre-print*

**2025** Symbolic Pauli Propagation for Gradient-Enabled Pre-Training of Quantum Circuits
S. **Monaco**, J. Slim, F. Rehm, D. Krücker, K. Borras

*arXiv pre-print*

**2025** Quantum Generative Modeling for Calorimeter Simulations on Noisy Quantum Devices
S. **Monaco**, J. Slim, F. Rehm, M. Scham, D. Krücker, K. Borras
*PoS* ACAT2025

**2025** Quantum Generative Modeling for Calorimeter Simulations on Noisy Quantum Devices
S. **Monaco**, J. Slim, F. Rehm, M. Scham, D. Krücker, K. Borras
*PoS* EPS-HEP2025 p. 639

**2024** Exploring the Phase Diagram of the quantum one-dimensional ANNNI model
M. Cea, M. Grossi, S. **Monaco**, E. Rico, L. Tagliacozzo, S. Vallecorsa
*arXiv pre-print*

**2024** Quantum Machine Learning for data analysis at LHCb
A. Gianelle, D. Lucchesi, S. **Monaco**, D. Nicotra, L. Sestini, D. Zuliani
*Il Nuovo Cimento C: colloquia and communications in physics* 47 (3) p. 127. Editrice Compositori srl

**2023** Quantum phase detection generalization from marginal quantum neural network models
S. **Monaco**, O. Kiss, A. Mandarino, S. Vallecorsa, M. Grossi
*Phys. Rev. B* 107 (8) p. L081105. American Physical Society

**2023** Study of quantum correlations in LHCb simulated heavy flavour jets
S. **Monaco**, D. Lucchesi, D. Zuliani, L. Sestini
*thesis.unipd.it*

## Talks

**2026** "Training of Large Shallow Parametrized Quantum Circuits through Pauli Propagation"
LHC Quantum Computing Workshop (University of Valencia), Valencia, Spain

**2026** "Training Quantum Generative Models for High Energy Physics"
FHSciComp 2026 (DESY), Hamburg, Germany

**2026** "Symbolic Pauli Propagation for Gradient-Enabled Pre-Training of Quantum Circuits"
CERN QTI Journal Club (CERN), Geneva, Switzerland

**2025** "Performance analysis of the QAG algorithm on QSEA and IBM quantum computers"
HQML Consortium Meeting, Hamburg, Germany

**2025** "Detector Simulation and Jet-Clustering for HL-LHC with Quantum Computing"
H2020 Engage Conference (ESRF), Grenoble, France

**2024** "Precise Quantum Angle Generator Designed for Noisy Quantum Devices"
Helmholtz Matter and the Universe Days 2024 (DESY), Hamburg, Germany

**2024** "Precise Quantum Angle Generator Designed for Noisy Quantum Devices"
FHSciComp 2024 (DESY), Hamburg, Germany

## Posters

**2026** "TBD"
QT4HEP 2026 (CERN), Geneva, Switzerland

**2026** "Symbolic Pauli Propagation for Gradient Enabled Pre-Training of Quantum Circuits"
QC4HEP 2026 (LMU Munich), Munich, Germany

**2025** "Scalable continuous-type generative model on superconducting quantum computers"
ACAT 2025 (University of Hamburg), Hamburg, Germany

**2025** "Precise Quantum Angle Generator Designed for Noisy Quantum Devices"
EPS 2025 (Palais du Pharo), Marseille, France

**2025** "Precise Quantum Angle Generator Designed for Noisy Quantum Devices"
QT4HEP 2025 (CERN), Geneva, Switzerland

**2024** "Precise Quantum Angle Generator Designed for Noisy Quantum Devices"
QTech 2024 (Freie Universität Berlin), Berlin, Germany

**2023** "Quantum Machine Learning for data analysis at LHCb"
IFAE 2023 (University of Catania), Catania, Italy

**2023** "Quantum phase detection generalization from marginal quantum neural network models"
QIP 2023 (Ghent University), Ghent, Belgium

## Extracurricular Activity

**Junction Quantum Hack 2026** — ***Espoo, Finland***

Hackathon Participant — *Jun .2026*

- Project: "Quantum Machine Learning for Finance".
- Goal: Implementation of a quantum algorithm to speed up derivative pricing beyond Black-Scholes.

### PennyLane Code Camp 2023 *Online*

Hackaton Participant *Nov. 2022*

- Team placed 7th out of 500+ teams in a coding challenge on Quantum Machine Learning with PennyLane.

### Computer Science Committee, College of Merit "Don Nicola Mazza" *Padua, Italy*

Lecturer, LaTeX Course *2021 - 2023*

- Taught a three-lesson course spanning fundamentals to advanced topics (bibliography, TikZ, custom commands).

## Writing

### PennyLane (Xanadu) *Online*

PennyLane Tutorial - Author *May 2025*

- Authored the tutorial "Supervised and Unsupervised Quantum Machine Learning Models for the Phase Detection of the ANNNI Spin Model" (pennylane.ai/qml/demos/tutorial_annni).

## Certificates

2021 **IELTS Academic: Score 7.0 (English, C1),** British Council

2016 **Goethe-Zertifikat B2 (German, B2),** Goethe-Institut

2016 **EsaBac Diploma (French-Italian dual diploma),** Ministère de l'Éducation nationale / MIUR